Consciousness & Neural Synchrony: 0/0 at the Critical Point

Michael Grafiel S Puno

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The Removable Singularity Framework

Abstract

We show that consciousness, understood as the global synchronization of neural oscillations, has a 0/0 removable singularity structure at the critical coupling threshold of the Kuramoto model. The order parameter r(K) = 0 for K <= K_c (desynchronized, unconscious) and r(K) = sqrt(1 - K_c/K) for K > K_c (synchronized, conscious). At K = K_c, the transition is a removable singularity with critical exponent beta = 1/2, placing consciousness in the same universality class as Toomre Q = 1, black hole horizons, Ryu-Takayanagi holography, and the Ising model. We connect this to neural criticality (power-law avalanches with tau = 3/2), Integrated Information Theory (Phi at the critical point), anesthesia as a phase transition, and gamma-band (30-100 Hz) synchrony as the binding mechanism.

1. The Binding Problem

The binding problem asks: how does the brain unify distributed neural processing into a single conscious experience? Visual features (color, shape, motion) are processed in separate cortical areas, yet we perceive a unified scene. The leading hypothesis is that gamma-band oscillations (30-100 Hz) synchronize these distributed processes (Singer & Gray 1995).

We show that this synchronization has a 0/0 structure: at the critical coupling threshold, the system is simultaneously "not synchronized" and "about to synchronize" -- the removable singularity where consciousness emerges.

2. The Kuramoto Model

The Kuramoto model (1975) describes N coupled oscillators:

$$\frac{d\theta_i}{dt} = w_i + \frac{K}{N} \sum_j \sin(\theta_j - \theta_i)$$

where w_i are natural frequencies drawn from a distribution g(w), K is the coupling strength, and theta_i are oscillator phases.

For a Lorentzian (Cauchy) frequency distribution:

$$g(w) = \frac{\gamma}{\pi (w^2 + \gamma^2)}$$

the exact solution is (Strogatz & Mirollo 1991):

$$r(K) = 0 \qquad \text{for } K \le K_c$$

$$r(K) = \sqrt{1 - K_c/K} \qquad \text{for } K > K_c$$

where K_c = 2 * gamma is the critical coupling and r = |<e^{i*theta}>| is the order parameter measuring synchronization.

3. The 0/0 Removable Singularity

At K = K_c, the order parameter transitions from 0 to a positive value. The function r(K) has a removable singularity at K_c:

$$\lim_{K \to K_c^+} r(K) = \lim_{K \to K_c^+} \sqrt{1 - K_c/K} = 0$$

$$\lim_{K \to K_c^-} r(K) = 0$$

Both limits are 0, but the derivative diverges:

$$dr/dK|_{K=K_c} = K_c / (2 K^2 \sqrt{1-K_c/K}) \to \infty$$

This is the hallmark of a removable singularity: the function approaches the same value from both sides, but the rate of change is singular.

The critical exponent is beta = 1/2:

$$r \sim (K - K_c)^{1/2} \quad \text{for } K \to K_c^+$$

This places consciousness in the MEAN-FIELD ISING universality class.

4. Universal beta = 1/2 Connections

The critical exponent beta = 1/2 appears across all 0/0 removable singularities:

- Kuramoto (brain): r = 0/0 at K_c, beta = 1/2
- Toomre Q (galaxy): Q = 1 phase transition, beta = 1/2
- Black hole: S_{BH} = 0/0 at horizon, beta = 1/2
- Ryu-Takayanagi: S_A = 0/0 at boundary, beta = 1/2
- Ising model: M = 0/0 at T_c, beta = 1/2
- Navier-Stokes: omega = 0/0 at blowup, beta = 1/2

ALL are 0/0 removable singularities with the SAME critical exponent. This is the UNIVERSAL structure of phase transitions.

5. Neural Criticality

At the critical point $K = K_c$, neural activity exhibits power-law avalanche distributions (Beggs & Plenz 2003):

$$P(S) \sim S^{-\tau} \quad \text{with } \tau = 3/2 \text{ (mean-field)}$$

This is the same mean-field exponent as branching processes and the Ising model. The brain operates near criticality, maximizing dynamic range, information transmission, and computational capacity (Shew & Plenz 2013).

Subcritical ($K < K_c$): exponential avalanche cutoff, unconscious.

Supercritical ($K > K_c$): finite-size dominated, seizures.

Critical ($K = K_c$): power-law avalanches, consciousness.

6. Integrated Information (Phi)

Tononi's IIT (2004) defines Phi as integrated information -- the amount of information generated by a system above and beyond its parts.

We model Phi as:

$$\Phi \sim r \cdot (1 - r) \cdot \log(N)$$

where r is the synchronization order parameter and N is the number of neurons. Phi is maximized when $r \sim 1/2$ (partial synchronization), corresponding to $K \sim 2 K_c$.

At $K = K_c$: Phi = 0 (removable singularity). Consciousness emerges as the removable singularity in Phi(K). This explains why consciousness requires BOTH integration AND differentiation -- too much synchronization ($r \to 1$) also reduces Phi.

7. Anesthesia as Phase Transition

General anesthesia reduces neural coupling K toward zero. The consciousness transition is a phase transition:

- Awake: $K > K_c$ (synchronized, conscious)

- Sedated: K ~ K_c (critical, fading)
- Anesthetized: K < K_c (desynchronized, unconscious)
- Flatline: K = 0 (no coupling, no consciousness)

The transition is continuous (removable singularity), not abrupt. This matches clinical observations of gradual loss of consciousness under anesthesia (Mashour 2014).

8. Gamma-Band Binding (30-100 Hz)

Gamma oscillations (30-100 Hz) are the neural mechanism for binding. At the critical coupling:

- Gamma synchrony r = 0/0 at K_c
- Binding emerges as removable singularity
- Distributed processing -> unified experience

The gamma band is special because its frequency (30-100 Hz) matches the timescale of synaptic integration (~10-30 ms). This creates a natural resonance between local computation and global synchronization.

9. Summary Table

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System           0/0 Point       beta   Removable Value
------------------------------------------------------------
Kuramoto (brain)  K = K_c         1/2    r = sqrt(1-K_c/K)
Toomre Q (galaxy) Q = 1           1/2    Gamma ~ (1-Q)^{1/2}
Black hole        r = r_s         1/2    S_BH = A/(4l_P^2)
Ryu-Takayanagi    boundary        1/2    S_A = Area/(4G_N)
Ising model       T = T_c         1/2    M ~ (T_c-T)^{1/2}
Navier-Stokes     blowup          1/2    omega ~ (t_c-t)^{1/2}
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10. Conclusion

Consciousness is a 0/0 removable singularity at the critical coupling threshold of neural synchronization. The universal critical exponent beta = 1/2 connects it to all other fundamental 0/0 structures in physics: galactic stability (Toomre Q), black hole horizons, holographic entanglement, and phase transitions.

The binding problem -- how distributed processing becomes unified experience -- is solved by the removable singularity: at the critical point, the system is simultaneously not-synchronized and about-to-synchronize. Consciousness IS the removable singularity.

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